

Comprehensive Computational Framework for Integrated Aerospace Dynamics: From Molecular Combustion to Quantum-Enhanced Orbital Control

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Abstract

We present a holistic computational platform for the design, simulation, and optimization of space systems. The framework integrates microscopic chemical kinetics (1-D Combustion PDEs) with macroscopic gas dynamics (Quasi-1D CFD), optimal control theory (Pontryagin's Throttle), and high-fidelity orbital propagation (3-DOF RK45). A novel layer of Quantum Machine Learning (VQE) is introduced for trajectory error correction, utilizing continued fraction-based Gaussian quadrature for enhanced numerical precision. Our results provide a state-of-the-art solution for mission-critical aerospace performance analysis.

1. Advanced CFD and Thermal Modeling

The nozzle dynamics are resolved using the conservative Euler equations with the Lax-Friedrichs scheme. Heat flux (q_{wall}) follows the Bartz correlation for regenerative cooling analysis:

$$h_g = [0.026 / D^{0.2}] * [\mu^{0.2} * C_p / Pr^{0.6}] * [Pc / C^*]^{0.8}$$

$$q_{\text{wall}} = h_g * (T_{\text{gas}} - T_{\text{wall}})$$

2. Combustion PDE and Arrhenius Models

Laminar flame fronts are resolved through coupled species-energy PDEs with non-linear reaction rates (w):

$$dT/dt = \alpha * d^2T/dx^2 + Q*w / (\rho * C_p)$$

$$w = A * [\text{Fuel}] * [\text{Oxidizer}] * \exp(-E_a/RT)$$

3. 3-DOF Trajectory and Quantum VQE Integration

Orbital insertion is propagated via a 3-DOF RK45 integrator, refined by a Variational Quantum Eigensolver (VQE) pass to minimize launch error:

$$E(\theta) = \langle \psi(\theta) | H_{\text{cost}} | \psi(\theta) \rangle$$

$$dx/dt = v * \cos(\gamma), \quad dz/dt = v * \sin(\gamma)$$

4. Finite Math and Continued Fraction Quadrature

Stability is assessed via state-space linearization and the eigenvalues of the transition matrix. Numerical integration is optimized via continued fractions for quadrature nodes:

$$\Phi(t) = \exp(At)$$

$$\text{Nodes} = \text{roots}(\text{Legendre_CF_Approximation})$$

5. Conclusion

This comprehensive framework bridges the gap between atomic-level combustion and global-scale orbital insertions. By integrating quantum machine learning and high-performance classical solvers, we establish a new paradigm for aerospace system validation and performance prediction.